

ASEN 6044

Homework #2

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Problem 1:

We are given a prior and measurement likelihood given by

$$P(\theta) = \begin{cases} \alpha e^{-\alpha\theta} & \theta \geq 0 \\ 0 & \theta < 0 \end{cases}$$

$$P(y | \theta) = \frac{\theta^y e^{-\theta}}{y!} \quad y \in \mathbb{Z}^+$$

From Bayes Rule, we know the posterior distribution is given by

$$P(\theta | y) = \frac{P(y | \theta)P(\theta)}{P(y)}$$

Where $P(y)$ is given by the marginalization of the joint distribution $P(\theta, y)$, that is

$$P(y) = \int_{-\infty}^{\infty} P(\theta, y) d\theta = \int_{-\infty}^{\infty} P(y | \theta) P(\theta) d\theta = \int_0^{\infty} \alpha e^{-\alpha\theta} \frac{\theta^y e^{-\theta}}{y!} d\theta$$

Since we are given y , $P(y)$ is a constant and is not dependant on θ . Thus, our posterior distribution will be proportional to the measurement likelihood times the prior.

$$P(\theta | y) \propto \alpha e^{-\alpha\theta} \frac{\theta^y e^{-\theta}}{y!} = \frac{\alpha \theta^y e^{-\theta(1+\alpha)}}{y!}$$

Recall the pdf of a gamma distribution:

$$f(x) = \Gamma(\gamma, \beta) = \frac{\beta^\gamma x^{\gamma-1} e^{-\beta x}}{\Gamma(\gamma)}$$

Where $\Gamma(\gamma)$ is the gamma function. For positive integers, the gamma function returns $\Gamma(n) = (n-1)!$. Since y is a positive integer, it's clear that our posterior distribution is a gamma distribution with parameters $\gamma = y + 1$ and $\beta = \alpha + 1$.

- a) The MMSE estimate for our posterior distribution is the expected value of the distribution.

$$\hat{\theta}_{MMSE} = \mathbb{E}[P(\theta | y)]$$

For a gamma distribution, the expected value is given by $\mathbb{E}[\Gamma(\gamma, \beta)] = \frac{\gamma}{\beta}$. Thus, the MMSE estimate for our posterior distribution is

$$\boxed{\hat{\theta}_{MMSE} = \frac{y+1}{\alpha+1}}$$

- b) The MAP estimate for our posterior distribution is the maximum value of our posterior. We can find the maximum by taking the derivative of our posterior with respect to θ and setting equal to 0.

$$\begin{aligned}\frac{d}{d\theta} \left[\frac{\alpha \theta^y e^{-\theta(1+\alpha)}}{y!} \right] &= \frac{\alpha [y \theta^{y-1} e^{-\theta(1+\alpha)} - (1+\alpha) e^{-\theta(1+\alpha)} \theta^y]}{y!} = 0 \\ [y \theta^{y-1} - (\alpha + 1) \theta^y] e^{-\theta(1+\alpha)} &= 0 \rightarrow (\alpha + 1) \theta^y = y \theta^{y-1} \\ \alpha + 1 = y \theta^{-1} &\rightarrow \boxed{\hat{\theta}_{MAP} = \frac{y}{\alpha + 1}}\end{aligned}$$

Problem 2:

We are given that a continuous random variable θ is uniformly distributed on the interval $[0, 1]$, and that measurements are given by $y = \theta + n$ where n 's distribution is given as

$$P(n) = \begin{cases} e^{-n} & n \geq 0 \\ 0 & n < 0 \end{cases}$$

From the following pdf, we can derive our measurement likelihood $P(y | \theta)$. Since our noise is additive, we have $n = y - \theta$. Plugging this into $P(n)$ we have

$$P(y | \theta) = P(y - \theta) = \begin{cases} e^{-y+\theta} & y \geq \theta \\ 0 & y < \theta \end{cases}$$

For a fixed observation y , this is identical to the measurement likelihood, $P(y | \theta)$. Applying Bayes Rule, we can derive an expression for our posterior.

$$P(\theta | y) = \frac{P(y | \theta) P(\theta)}{\int_{-\infty}^{\infty} P(y | \theta) P(\theta) d\theta}$$

The bounds on the posterior must be consistent with the constraints placed on the prior and likelihood. We know $\theta \in [0, 1]$ and $y \geq \theta$. Thus, we know our posterior estimate should be cut off at $\theta = 1$ if $y \geq 1$. However if $y < 1$, we know that θ cannot be larger than y . Thus an upperbound on our posterior distribution becomes $u = \min(1, y)$. We can use this upper bound to solve for the normalizing integral in posterior expression.

$$\begin{aligned}\int_{-\infty}^{\infty} P(y | \theta) P(\theta) d\theta &= \int_0^u P(y | \theta) P(\theta) d\theta = \int_0^u e^{-y} e^{\theta} d\theta \\ &= e^{-y} [e^u - 1]\end{aligned}$$

And our posterior becomes

$$P(\theta | y) = \begin{cases} \frac{e^{\theta}}{e^u - 1} & 0 \leq \theta \leq u \\ 0 & \text{o.w} \end{cases} \quad u = \min(1, y)$$

- a) $\hat{\theta}_{MMSE} = \mathbb{E}[P(\theta | y)]$ by definition. Thus we can evaluate the expected value to receive an expression for $\hat{\theta}_{MMSE}$.

$$\begin{aligned}\mathbb{E}[P(\theta | y)] &= \int_0^u \theta \frac{e^{\theta}}{e^u - 1} d\theta = \frac{1}{e^u - 1} \left[\theta e^{\theta} \Big|_0^u - \int_0^u e^{\theta} d\theta \right] \\ &= \frac{1}{e^u - 1} [u e^u - e^u + 1] = \frac{u e^u - e^u + 1}{e^u - 1}\end{aligned}$$

Thus $\hat{\theta}_{MMSE} = \mathbb{E}[P(\theta | y)] = \boxed{\frac{ue^u - e^u + 1}{e^u - 1}}$

b) By taking the derivative of our posterior we can derive an expression for $\hat{\theta}_{MAP}$.

$$\frac{d}{d\theta} \left[\frac{e^\theta}{e^u - 1} \right] = \frac{e^\theta}{e^u - 1}$$

Since this function is monotonic increasing, it's maximum must lie at the end of the interval.

Thus, $\boxed{\hat{\theta}_{MAP} = \min(u, 1)}$.

Problem 3:

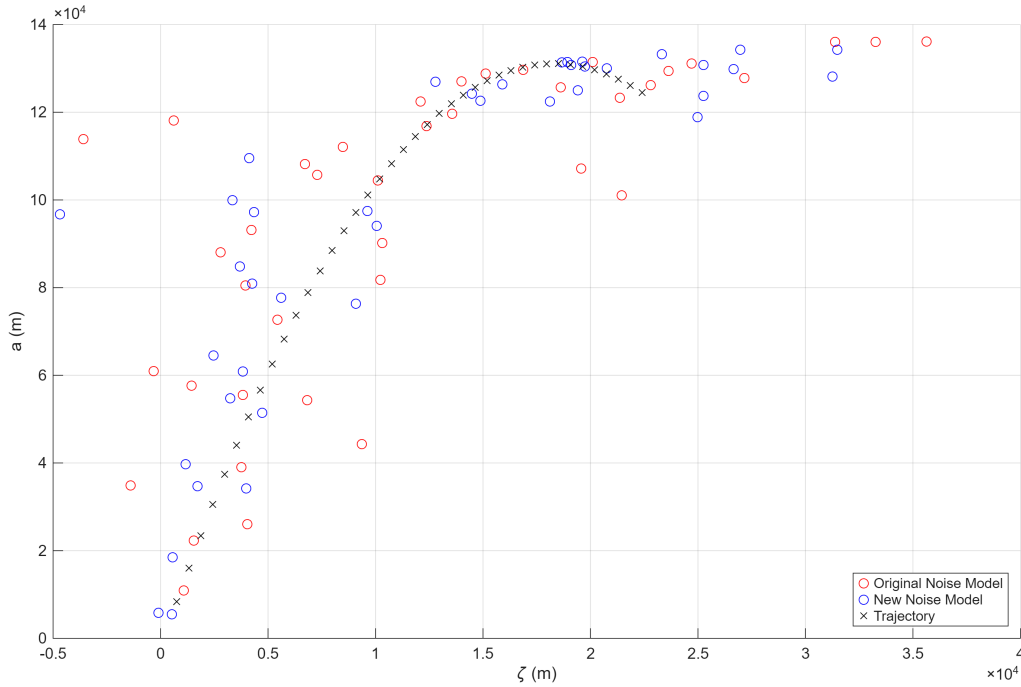


Figure 1: Comparison of Old and New Noisy Measurements

- From Figure 1 we see generally that the measurements from the new noisy range model draw out a similar shape to that of the old range model. Without any prior knowledge regarding the problem, it even appears that both sets of measurements have the same noise model.
- TODO: Best x_0 , Variance of range? Is it even supposed to work??
- The negative log-likelihood function, $\mathcal{L}(x_0; Y_{1:N}) = J_{NLL}$ is defined as the negative logarithm of the measurement likelihood functions, that is

$$J_{NLL} = -\log(P(Y_{1:N} | x_0))$$

To derive an expression for $P(Y_{1:N} | x_0)$, we look at the given measurement model:

$$\rho_k = \sqrt{(L - \xi(t_k))^2 + (a(t_k))^2} + v_{r,k} \quad v_{r,k} \sim \mathcal{T}_{v_{r,k}}\left(0, \nu = \frac{1}{2}\right) \quad (1)$$

$$\theta_k = \arctan\left(\frac{a(t_k)}{L - \xi(t_k)}\right) + v_{b,k} \quad v_{b,k} \sim \mathcal{N}_{v_{b,k}}\left(0, \sigma_b^2 = 0.005\right) \quad (2)$$

Where $v_{r,k}$ and $v_{b,k}$ are the range and bearing noise respectively. To derive the measurment likelihood $P(Y_{1:N} | x_0)$ we assume that the measurments are conditionally independent given x_0 , that is $P(Y_{1:N} | x_0) = \prod_{k=1}^N P(y_k | x_0)$. We also assume that the range and bearing measurments are independent of each other, our expression further simplifies to the following:

$$P(Y_{1:N} | x_0) = \prod_{k=1}^N P(\rho_k | x_0) P(\theta_k | x_0) \quad (3)$$

Since our noise is additive, solving for the respective probabilities is fairly straightforward. From (1) we can derive $P(\rho_k | x_0)$ as follows:

$$v_{r,k} = \rho_k - \sqrt{(L - \xi(t_k))^2 + (a(t_k))^2}$$

$$P(v_{r,k}) = \frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2})\sqrt{\nu\pi}} \left(1 + \frac{v_{r,k}^2}{\nu}\right)^{-\frac{\nu+1}{2}}$$

Substituting for $v_{r,k}$ we get the following expression in terms of our initial conditions.

$$P(\rho_k | x_0) = \frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2})\sqrt{\nu\pi}} \left(1 + \frac{(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t_k)^2 + (a_0 + \dot{a}_0 t_k - 0.5gt_k^2)})^2}{\nu}\right)^{-\frac{\nu+1}{2}} \quad (4)$$

We can follow a similar process to derive $P(\theta_k | x_0)$.

$$v_{b,k} = \theta_k - \arctan\left(\frac{a(t_k)}{\xi(t_k)}\right)$$

$$P(v_{b,k}) = \frac{1}{\sigma_b\sqrt{2\pi}} \exp\left\{\frac{-1}{2\sigma_b^2} v_{b,k}^2\right\}$$

$$P(\theta_k | x_0) = \frac{1}{\sigma_b\sqrt{2\pi}} \exp\left\{\frac{-\left[\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right)\right]^2}{2\sigma_b^2}\right\} \quad (5)$$

Substituting (5) and (4) into (3) and taking the negative logarithm we recieve the following

expression for our negative log likelihood function:

$$\begin{aligned}
\mathcal{L}(x_0; Y_{1:N}) &= - \sum_{k=1}^N \log[P(\rho_k | x_0)P(\theta_k | x_0)] \\
&= - \sum_{k=1}^N \log \left(\frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2})\sqrt{\nu\pi}} \frac{1}{\sigma_b\sqrt{2\pi}} \right) \\
&\quad - \sum_{k=1}^N -\frac{\nu+1}{2} \log \left(1 + \frac{(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t_k)^2 + (a_0 + \dot{a}_0 t_k - 0.5gt_k^2)})^2}{\nu} \right) \\
&\quad - \sum_{k=1}^N \frac{\left[\theta_k - \arctan \left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right) \right]^2}{2\sigma_b^2}
\end{aligned} \tag{6}$$

Both the NLL and NLS cost functions seek to find an x_0 that minimizes the cost function given observations $Y_{1:N}$, however they each approach the problem in a different way. NLS is defined to minimize the sum of the squared residuals, while maximum likelihood tries to find the x_0 which makes ‘the most sense’ given all the observations. Both assume the measurement noise characteristics are known, NLS uses a least squares cost function, meaning we only consider the 1st/2nd order moments. As we have seen with the Student’s-t distribution, this does not accurately capture higher order moments. Maximum likelihood is robust to non-Gaussian sensor noise unlike the least-squares cost function we have selected for NLS.

d) In order to minimize (6), we take the gradient with respect to x_0 , or $\nabla_{x_0} \mathcal{L}(x_0; Y_{1:N})$, where

$$\nabla_{x_0} = \begin{bmatrix} \frac{\partial}{\partial \xi_0} \\ \frac{\partial}{\partial \dot{\xi}_0} \\ \frac{\partial}{\partial a_0} \\ \frac{\partial}{\partial \dot{a}_0} \end{bmatrix}$$

$$\nabla_{x_0} \mathcal{L}(x_0; Y_{1:N}) = -\nabla_{x_0} \sum_{k=1}^N \log[P(\rho_k | x_0)P(\theta_k | x_0)]$$

It’s easy to show that the expression simplifies to the following, where the constant terms evaluate to zero.

$$\begin{aligned}
&= -\nabla_{x_0} \left[\frac{-\nu+1}{2} \log \left(1 + \frac{(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t_k)^2 + (a_0 + \dot{a}_0 t_k - 0.5gt_k^2)})^2}{\nu} \right) \right] \\
&\quad - \nabla_{x_0} \left[\frac{-1}{2} \left[\frac{\theta_k - \arctan \left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k} \right)}{\sigma_b} \right]^2 \right]
\end{aligned} \tag{7}$$

Equation (7) consists of two gradient terms to evaluate. Starting with the first term, we can take each partial derivative independently and stack the result into a vector:

$$\nabla_{x_0} \left[\frac{-\nu+1}{2} \log \left(1 + \frac{\left(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t_k)^2 + (a_0 + \dot{a}_0 - 0.5gt_k^2)^2} \right)^2}{\nu} \right) \right]$$

$$\begin{aligned} & \frac{\partial}{\partial \xi_0} \left[\frac{-\nu+1}{2} \log \left(1 + \frac{\left(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t_k)^2 + (a_0 + \dot{a}_0 - 0.5gt_k^2)^2} \right)^2}{\nu} \right) \right] \\ &= \frac{2(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt_k^2)^2})(L - \xi_0 - \dot{\xi}_0 t)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_o + \dot{a}_0 t - 0.5gt^2)^2} \right)^2 + \nu \right) \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_o + \dot{a}_0 t - 0.5gt^2)^2}} \end{aligned}$$

$$\begin{aligned} & \frac{\partial}{\partial \xi_0} \left[\frac{-\nu+1}{2} \log \left(1 + \frac{\left(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t_k)^2 + (a_0 + \dot{a}_0 - 0.5gt_k^2)^2} \right)^2}{\nu} \right) \right] \\ &= \frac{2t(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt_k^2)^2})(L - \xi_0 - \dot{\xi}_0 t)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_o + \dot{a}_0 t - 0.5gt^2)^2} \right)^2 + \nu \right) \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_o + \dot{a}_0 t - 0.5gt^2)^2}} \end{aligned}$$

$$\begin{aligned} & \frac{\partial}{\partial a_0} \left[\frac{-\nu+1}{2} \log \left(1 + \frac{\left(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t_k)^2 + (a_0 + \dot{a}_0 - 0.5gt_k^2)^2} \right)^2}{\nu} \right) \right] \\ &= \frac{-2(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt_k^2)^2})(a_0 + \dot{a}_0 t - 0.5gt^2)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_o + \dot{a}_0 t - 0.5gt^2)^2} \right)^2 + \nu \right) \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_o + \dot{a}_0 t - 0.5gt^2)^2}} \end{aligned}$$

$$\begin{aligned} & \frac{\partial}{\partial \dot{a}_0} \left[\frac{-\nu+1}{2} \log \left(1 + \frac{\left(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t_k)^2 + (a_0 + \dot{a}_0 - 0.5gt_k^2)^2} \right)^2}{\nu} \right) \right] \\ &= \frac{-2t(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt_k^2)^2})(a_0 + \dot{a}_0 t - 0.5gt^2)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_o + \dot{a}_0 t - 0.5gt^2)^2} \right)^2 + \nu \right) \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_o + \dot{a}_0 t - 0.5gt^2)^2}} \end{aligned}$$

Therefore,

$$\nabla_{x_0} \left[\frac{-\nu+1}{2} \log \left(1 + \frac{\left(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0 t_k)^2 + (a_0 + \dot{a}_0 - 0.5gt_k^2)^2} \right)^2}{\nu} \right) \right]$$

$$= \frac{-\nu + 1}{2} \left[\frac{\frac{2(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt_k^2)^2})(L - \xi_0 - \dot{\xi}_0 t)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt^2)^2}\right)^2 + \nu\right)\sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt^2)^2}}}{\frac{2t(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt_k^2)^2})(L - \xi_0 - \dot{\xi}_0 t)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt^2)^2}\right)^2 + \nu\right)\sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt^2)^2}}} - \frac{-2(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt_k^2)^2})(a_0 + \dot{a}_0 t - 0.5gt^2)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt^2)^2}\right)^2 + \nu\right)\sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt^2)^2}}}{\frac{-2t(\rho_k - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt_k^2)^2})(a_0 + \dot{a}_0 t - 0.5gt^2)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt^2)^2}\right)^2 + \nu\right)\sqrt{(L - \xi_0 - \dot{\xi}_0)^2 + (a_0 + \dot{a}_0 t - 0.5gt^2)^2}}} \right] \quad (8)$$

We will follow a similar process for the second gradient term.

$$\nabla_{x_0} \left[\frac{-1}{2} \left[\frac{\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right)}{\sigma_b} \right]^2 \right] = \frac{-1}{2\sigma_b^2} \nabla_{x_0} \left[\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right) \right]^2$$

$$\frac{\partial}{\partial \xi_0} \left[\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right) \right]^2 = \frac{-2t[\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right)](a_0 + \dot{a}_0 t_k - 0.5gt_k^2)}{\left(\left[\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right]^2 + 1\right)(L - \xi_0 - \dot{\xi}_0 t_k)^2}$$

$$\frac{\partial}{\partial \dot{\xi}_0} \left[\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right) \right]^2 = \frac{-2[\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right)](a_0 + \dot{a}_0 t_k - 0.5gt_k^2)}{\left(\left[\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right]^2 + 1\right)(L - \xi_0 - \dot{\xi}_0 t_k)^2}$$

$$\frac{\partial}{\partial a_0} \left[\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right) \right]^2 = \frac{-2[\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right)]}{\left(\left[\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right]^2 + 1\right)(L - \xi_0 - \dot{\xi}_0 t_k)}$$

$$\frac{\partial}{\partial \dot{a}_0} \left[\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right) \right]^2 = \frac{-2t[\theta_k - \arctan\left(\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right)]}{\left(\left[\frac{a_0 + \dot{a}_0 t_k - 0.5gt_k^2}{L - \xi_0 - \dot{\xi}_0 t_k}\right]^2 + 1\right)(L - \xi_0 - \dot{\xi}_0 t_k)}$$

Therefore,

$$\nabla_{x_0} \left[\frac{-1}{2} \left[\frac{\theta_k - \arctan\left(\frac{a_0 + a_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \xi_0 t_k}\right)}{\sigma_b} \right]^2 \right] = \frac{-1}{2\sigma_b^2} \left[\begin{array}{c} -2[\theta_k - \arctan\left(\frac{a_0 + a_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \xi_0 t_k}\right)](a_0 + a_0 t_k - 0.5 g t_k^2) \\ \frac{([\frac{a_0 + a_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \xi_0 t_k}]^2 + 1)(L - \xi_0 - \xi_0 t_k)^2}{-2[\theta_k - \arctan\left(\frac{a_0 + a_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \xi_0 t_k}\right)](a_0 + a_0 t_k - 0.5 g t_k^2)} \\ \frac{([\frac{a_0 + a_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \xi_0 t_k}]^2 + 1)(L - \xi_0 - \xi_0 t_k)^2}{-2[\theta_k - \arctan\left(\frac{a_0 + a_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \xi_0 t_k}\right)]} \\ \frac{([\frac{a_0 + a_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \xi_0 t_k}]^2 + 1)(L - \xi_0 - \xi_0 t_k)}{-2t[\theta_k - \arctan\left(\frac{a_0 + a_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \xi_0 t_k}\right)]} \\ \frac{([\frac{a_0 + a_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \xi_0 t_k}]^2 + 1)(L - \xi_0 - \xi_0 t_k)}{-2t[\theta_k - \arctan\left(\frac{a_0 + a_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \xi_0 t_k}\right)]} \end{array} \right] \quad (9)$$

Finally, we can substitute equations (8) and (9) into equation (7) and we are left with an expression for the gradient vector of $\mathcal{L}_{NLL}$

$$\nabla_{x_0} \mathcal{L}(x_0; Y_{1:N}) = \sum_{k=1}^N \left[\frac{\nu - 1}{2} \left[\begin{array}{c} \frac{2(\rho_k - \sqrt{(L - \xi_0 - \xi_0 t)^2 + (a_0 + a_0 t - 0.5 g t^2)^2})(L - \xi_0 - \xi_0 t)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \xi_0 t)^2 + (a_0 + a_0 t - 0.5 g t^2)^2} \right)^2 + \nu \right) \sqrt{(L - \xi_0 - \xi_0 t)^2 + (a_0 + a_0 t - 0.5 g t^2)^2}} \\ \frac{2t(\rho_k - \sqrt{(L - \xi_0 - \xi_0 t)^2 + (a_0 + a_0 t - 0.5 g t^2)^2})(L - \xi_0 - \xi_0 t)}{\left(\left(\rho - \sqrt{(L - \xi_0 - \xi_0 t)^2 + (a_0 + a_0 t - 0.5 g t^2)^2} \right)^2 + \nu \right) \sqrt{(L - \xi_0 - \xi_0 t)^2 + (a_0 + a_0 t - 0.5 g t^2)^2}} \\ -2(\rho_k - \sqrt{(L - \xi_0 - \xi_0 t)^2 + (a_0 + a_0 t - 0.5 g t^2)^2})(a_0 + a_0 t - 0.5 g t^2) \\ \frac{\left(\left(\rho - \sqrt{(L - \xi_0 - \xi_0 t)^2 + (a_0 + a_0 t - 0.5 g t^2)^2} \right)^2 + \nu \right) \sqrt{(L - \xi_0 - \xi_0 t)^2 + (a_0 + a_0 t - 0.5 g t^2)^2}}{-2t(\rho_k - \sqrt{(L - \xi_0 - \xi_0 t)^2 + (a_0 + a_0 t - 0.5 g t^2)^2})(a_0 + a_0 t - 0.5 g t^2)} \\ \frac{\left(\left(\rho - \sqrt{(L - \xi_0 - \xi_0 t)^2 + (a_0 + a_0 t - 0.5 g t^2)^2} \right)^2 + \nu \right) \sqrt{(L - \xi_0 - \xi_0 t)^2 + (a_0 + a_0 t - 0.5 g t^2)^2}}{-2t(\rho_k - \sqrt{(L - \xi_0 - \xi_0 t)^2 + (a_0 + a_0 t - 0.5 g t^2)^2})(a_0 + a_0 t - 0.5 g t^2)} \end{array} \right] \\ + \frac{1}{2\sigma_b^2} \left[\begin{array}{c} -2[\theta_k - \arctan\left(\frac{a_0 + a_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \xi_0 t_k}\right)](a_0 + a_0 t_k - 0.5 g t_k^2) \\ \frac{([\frac{a_0 + a_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \xi_0 t_k}]^2 + 1)(L - \xi_0 - \xi_0 t_k)^2}{-2t[\theta_k - \arctan\left(\frac{a_0 + a_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \xi_0 t_k}\right)](a_0 + a_0 t_k - 0.5 g t_k^2)} \\ \frac{([\frac{a_0 + a_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \xi_0 t_k}]^2 + 1)(L - \xi_0 - \xi_0 t_k)^2}{-2[\theta_k - \arctan\left(\frac{a_0 + a_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \xi_0 t_k}\right)]} \\ \frac{([\frac{a_0 + a_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \xi_0 t_k}]^2 + 1)(L - \xi_0 - \xi_0 t_k)}{-2t[\theta_k - \arctan\left(\frac{a_0 + a_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \xi_0 t_k}\right)]} \\ \frac{([\frac{a_0 + a_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \xi_0 t_k}]^2 + 1)(L - \xi_0 - \xi_0 t_k)}{-2t[\theta_k - \arctan\left(\frac{a_0 + a_0 t_k - 0.5 g t_k^2}{L - \xi_0 - \xi_0 t_k}\right)]} \end{array} \right] \quad (10)$$

e)

f)

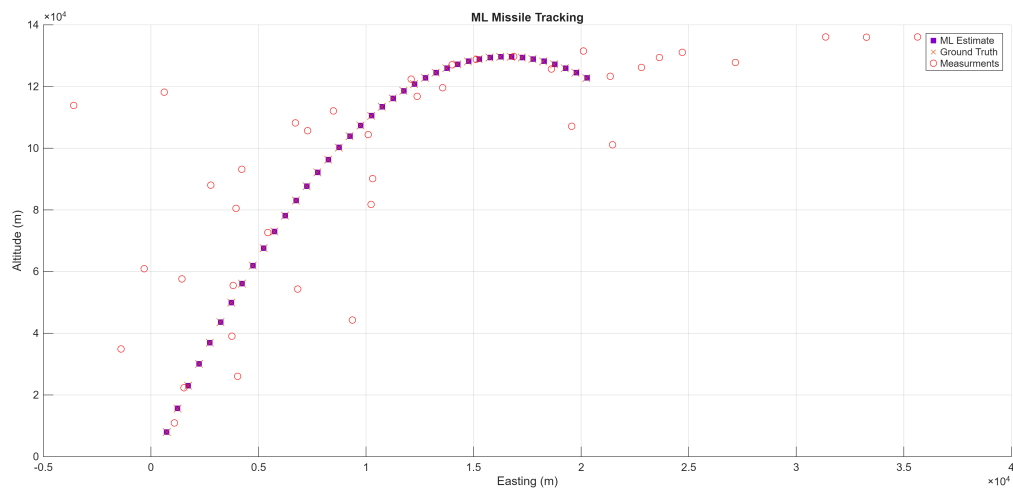


Figure 2: Maximum Likelihood Estimated Trajectory